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Since Equity Value is based on Share Price * Share Count, and neither component can be negative, *effectively*, it cannot be negative (the Implied Equity Value from a valuation methodology could still be negative, but you typically set it to \$0 when this happens).

Enterprise Value *could* be negative since Cash could exceed Equity Value for certain types of companies, such as “busted biotech” firms that trade at a discount to their cash.

12. You are comparing Companies A and B. Each operates in the same industry with the same revenue, EBITDA, and other financial metrics.

Company A is financed with 100% Equity, and Company B is financed with 50% Equity and 50% Debt.

In theory, their Enterprise Values should be the same. Will they be the same in real life?

No, most likely not. Although people claim that Enterprise Value is “capital structure-neutral,” it’s more accurate to say that it is *less affected* by capital structure than Equity Value.

The specific issue is that a company’s capital structure affects its **Discount Rate** because a shifting Debt and Equity mix changes its risk and potential returns.

Using more Debt increases the company’s default risk, which affects *all the investors* – even the common shareholders.

So, if investors value these companies based on their future cash flows and separate [Discount Rates \(WACC\)](#), the Discount Rate differential could explain valuation differences.

The companies’ “Current” Enterprise Values might still be close, but you will see more of a difference with the “Implied” versions, which may grow over time.

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Equity Value & Enterprise Value – Calculations

These questions are simple if you remember the two key rules:

1) Does Common Shareholders’ Equity (CSE) change?

If so, then Equity Value changes by the amount that CSE changes. If not, then Equity Value does not change.



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Items that affect CSE include Net Income, Dividends, Stock Issuances, and Stock Repurchases.

2) Do **Net Operating Assets (NOA)** change?

If so, then Enterprise Value changes by the amount that NOA changes. It doesn't matter which investor group was responsible because Enterprise Value reflects all investors.

The questions and answers here use the following abbreviations and assumptions:

- TEV = Enterprise Value
- Eq Val = Equity Value
- CSE = Common Shareholders' Equity
- NOA = Net Operating Assets
- Tax Rate = 25%

1. A company issues \$200 in Common Shares. How do Equity Value and Enterprise Value change?

CSE increases by \$200, so Eq Val increases by \$200.

NOA does not change because neither Cash nor CSE is operational, so TEV stays the same.

Alternatively, in the TEV formula, the extra Cash offsets the higher Equity Value.

2. This same company decides to use the \$200 in Common Stock proceeds to acquire another business for \$100 instead. How does everything change?

CSE increases by \$200 from this issuance, so Eq Val increases by \$200.

Of this \$200 in proceeds, \$100 remains in Cash, and \$100 is allocated to Acquired Assets from the other business.

These Acquired Assets are Operating Assets, and no Operating Liabilities change, so **NOA** increases by \$100. TEV, therefore, increases by \$100.

3. What if the company uses that same \$100 from the \$200 of new Common Stock to acquire an Asset rather than an entire company?

CSE still increases by \$200, so Eq Val is up by \$200.



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If this Asset is considered “Operating” or “Core,” such as a factory, then NOA increases by \$100, so TEV also increases by \$100.

If not – for example, the Asset is a short-term investment – then NOA does not change, and TEV stays the same.

4. A company issues \$100 in Debt to purchase a new factory. How do Equity Value and Enterprise Value change?

CSE does not change because Debt, not Equity, is the financing source here. Therefore, Eq Val stays the same.

The new factory is a Net Operating Asset, so NOA increases by \$100, and TEV, therefore, also increases by \$100.

Alternatively, in the TEV formula, Equity Value stays the same, and Debt increases, so TEV goes up.

5. A company issues \$100 of Common Stock and \$100 of Preferred Stock and lets the proceeds sit in Cash. How do Equity Value and Enterprise Value change?

CSE increases by \$100 due to the Common Stock issuances, so Eq Val is up by \$100. The Preferred Stock issuance does **not** affect Common Shareholders’ Equity!

Net Operating Assets are unchanged, so TEV stays the same.

In the TEV formula, Equity Value is up by \$100, Cash is \$200 more negative, and Preferred Stock is up by \$100, so TEV stays the same.

6. This same company now issues \$10 in Common Dividends and \$10 in Preferred Dividends. What happens in JUST THIS STEP?

CSE decreases by \$20 due to these Dividend issuances. Yes, both Common and Preferred Dividends flow into Common Shareholders’ Equity on the Balance Sheet. Therefore, Eq Val is down by \$20.

Net Operating Assets are unchanged, so TEV stays the same.



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7. A company issues \$150 of Debt and \$50 of Common Stock to acquire \$175 of PP&E and \$25 of Short-Term Investments. How do Equity Value and Enterprise Value change?

CSE increases by \$50 because of the Common Stock Issuance, so Eq Val increases by \$50.

The \$175 of PP&E counts as an Operating Asset, and no Operating Liabilities change, so **NOA** increases by \$175, and TEV also increases by \$175.

8. A company issues \$50 of Debt to buy a new factory. However, AFTER this purchase, its Enterprise Value increases by \$100 rather than \$50.

Your co-worker claims it is because the company issued Debt to make this purchase, and Debt increases Enterprise Value. Is he correct?

No. The *purchase method* does not matter when determining how Enterprise Value changes. All that matters is how the Net Operating Assets change.

The most likely explanation here is that the *book value* of this factory is \$50, but its *market value* is \$100 because market participants believe the [Present Value of its future cash flows](#) is closer to \$100. Therefore, Enterprise Value increases by \$100 rather than \$50. The company effectively got a discount on a new asset.

9. A company purchases \$100 of Inventory using Cash. How do Equity Value and Enterprise Value change?

There are no changes on the Income Statement in this initial step because the Inventory has not yet been sold. Therefore, Net Income does not change.

On the Balance Sheet, **CSE** stays the same in this initial step (no changes to Net Income, Stock Issuances/Repurchases, Dividends, etc.), so Eq Val stays the same.

NOA increases by \$100 since Inventory is an Operating Asset, and no Operating Liabilities change, so TEV increases by \$100.

10. Now assume the Inventory is sold for \$200 and walk me through how the *entire* process from beginning to end affects Equity Value and Enterprise Value.



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On the Income Statement, Revenue is up by \$200, and Pre-Tax Income is up by \$100 (due to the \$100 of Inventory now being recognized as COGS). Net Income increases by \$75 at a 25% tax rate.

On the CFS, Net Income is up by \$75, and there are no other net changes (Inventory went up by \$100 and then went down by \$100), so Cash is up by \$75 at the bottom.

On the Balance Sheet, Cash is up by \$75 on the Assets side, and CSE is up by \$75 on the L&E side.

Since **CSE** is up by \$75, Eq Val increases by \$75.

NOA does not change because Cash is not an Operating Asset, and no Operating Liabilities change, so TEV stays the same.

Intuition: This 2-step process represents the company generating Net Income and letting it sit in Cash; that process does not make its core business more valuable, so TEV does not increase.

11. A company has 200 outstanding shares at a current price of \$10.00. It also has 50 options at an exercise price of \$8.00 each. What is its Diluted Equity Value?

The Basic Equity Value is $200 * \$10.00 = \$2,000$. To calculate the diluted shares, note that the options are all “in the money” – their exercise price is less than the current share price – and apply the Treasury Stock Method.

When these options are exercised, 50 new shares are created.

The investors paid the company \$8.00 to exercise each option, so the company gets \$400 in cash. It uses that cash to buy back $\$400 / \$10.00 = 40$ of the new shares, so the diluted share count is $200 + 50 - 40 = 210$, and the Diluted Equity Value is \$2,100.

12. A company has 10,000 shares outstanding and a current share price of \$20.00. It also has 100 options at an exercise price of \$10.00, 50 Restricted Stock Units (RSUs), and 100 convertible bonds at a conversion price of \$10.00 and a par value of \$100.

What is its Diluted Equity Value?

For this type of question, **you should ask to write down the numbers.**

Since the options are in-the-money, you assume they get exercised, so 100 new shares are created.